Sindh Science Learning Portal

Class 5 · Science

Chapter 5

Matter and Its Changes



Learn what matter is, its three states, and the difference between physical and chemical changes.

Following the Sindh Textbook Board curriculum

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CHAPTER 5

Matter and Its Changes

What we will learn Learn what matter is, its three states, and the difference between physical and chemical changes.

1 What is matter?

Anything that **takes up space** and has **mass (weight)** is called **matter**.

Books, water, air, stones and our own bodies — all are matter.

We cannot see air, but it is matter too. When you blow into a balloon it gets bigger, because air takes up space.

Matter is made of tiny **particles** that are too small to see with our eyes.

2 The three states of matter

Matter is found in three states:

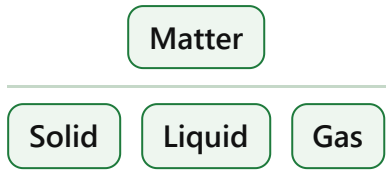
- **Solid**: has a fixed shape and a fixed size. Its particles are packed very close together. Examples: stone, wood, ice.
- **Liquid**: has a fixed size but no fixed shape. It takes the shape of the container it is poured into. Examples: water, milk, oil.
- **Gas**: has no fixed shape and no fixed size. It spreads out to fill all the space. Examples: air, steam, oxygen.

Heating or cooling can change matter from one state to another.



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Diagram: the parts and what they do



Matter (مادو · ماده)
Anything that takes up space and has mass.

Solid (ٺوس · ٹھوس)
Fixed shape and size. Particles packed close. Examples: stone, wood, ice.

Liquid (مائع · مائع)
Fixed size but no fixed shape. Takes the shape of its container. Examples: water, milk, oil.

Gas (گئس · گیس)
No fixed shape or size. Spreads to fill all the space. Examples: air, steam.

3 Physical and chemical changes

Matter can go through two kinds of change:

- **Physical change**: no new substance is made. Often the change can be reversed. Examples: ice melting, water turning into vapour, paper tearing, sugar dissolving in water.
- **Chemical change**: a **new substance** is made and the change cannot easily be reversed. Examples: paper burning, an egg cooking, iron getting **rust**, milk turning into yoghurt.

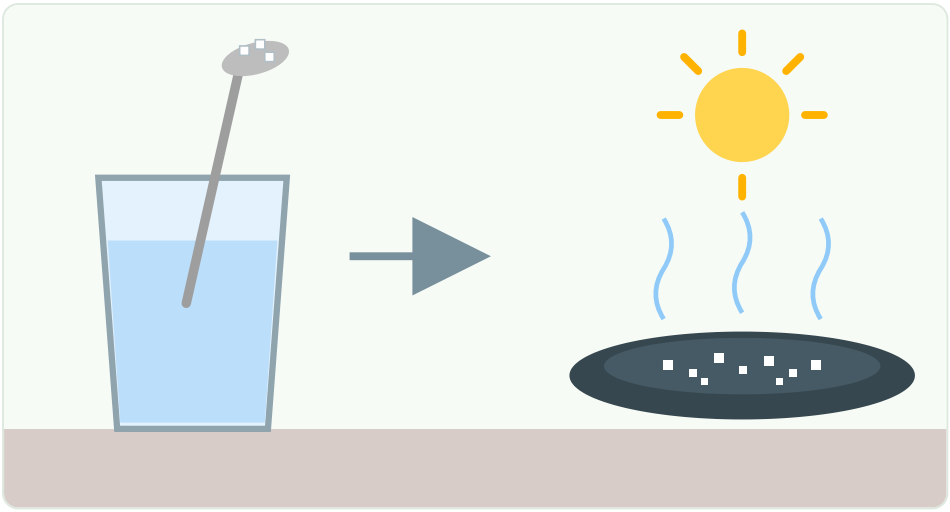
In a chemical change the colour often changes, heat or light is given out, or a new smell appears.

Key words

English	سنڌي	اردو
matter	مادو	ماده
mass	ڪميت	کمیت
particles	ذرڙا	ذرات
solid	ٺوس	ٹھوس
liquid	مائع	مائع
gas	گئس	گیس
physical change	طبعي تبديلي	طبعی تبدیلی
chemical change	ڪيميائي تبديلي	کیمیائی تبدیلی
dissolve	ڳرڻ (حل ٿيڻ)	گھلنا

English	سنڌي	اردو
rust	ڪٽ	زنگ

Experiment: Can we get dissolved salt back?



You need

- One spoon of salt
- Half a glass of water
- A dark or deep plate
- A spoon

Steps

1. Put the salt in the water and stir. The salt will disappear from sight.
2. Pour a little of the salty water into the plate.
3. Keep the plate in the sun for 2 or 3 days.
4. When all the water has dried up, look at the plate.

Safety: Do not drink the salty water, and wash your hands after the experiment.

What did we learn? White grains of salt are left on the plate. The water evaporated but the salt stayed the same. Salt dissolving in water is a physical change, because no new substance was made and we got the salt back.

Practice questions

Circle the correct answer

1. **What is matter?**

   (a) Anything that takes up space and has mass

   (b) Only things we can see

   (c) Only living things

   (d) Only solid things

2. **Why does a balloon get bigger when we blow into it?**

   (a) Air takes up space

   (b) The balloon is alive

   (c) The balloon melts

   (d) Air has no mass

3. **Which state has both a fixed shape and a fixed size?**

(a) Solid **(b)** Liquid

(c) Gas **(d)** None

4. **Milk takes the shape of the glass it is poured into. Which state is milk in?**

(a) Solid **(b)** Liquid

(c) Gas **(d)** Particle

5. **Which of these is a gas?**

(a) Wood **(b)** Oil

(c) Oxygen **(d)** Stone

6. **Which of these is a chemical change?**

(a) Ice melting **(b)** Paper tearing

(c) Paper burning **(d)** Sugar dissolving in water

7. **Which of these is a physical change?**

(a) An egg cooking **(b)** Iron rusting

(c) Water freezing into ice **(d)** Wood burning

8. **What happens in a chemical change?**

(a) A new substance is made **(b)** Only the shape changes

(c) Nothing happens **(d)** The thing always goes back to how it was

9. **Matter is made of tiny what?**

(a) Particles **(b)** Leaves

(c) Drops **(d)** Seeds

Write true or false

10. **Air is not matter because we cannot see it.** _____

11. **A gas spreads out to fill all the space.** _____

12. **In a solid, the particles are very far apart.** _____

13. **A cooked egg cannot be turned back into a raw egg.** _____

14. **Iron rusting is a physical change.** _____

Match the columns

15. **Match each thing with its state.**

1. Wood		(a) Liquid
2. Oil		(b) Gas
3. Steam		(c) Solid

Match the columns

16. **Match each change with its kind.**

1. Paper tearing		(a) Chemical change
2. Milk turning into yoghurt		(b) Physical change

Worksheet: Matter and Its Changes

Name: _____ Date: _____

A. Fill in the blanks

Word box: container · matter · new · rust · solid

1. Anything that takes up space and has mass is called _____ .
2. A _____ has a fixed shape and a fixed size.
3. A liquid takes the shape of its _____ .
4. In a chemical change a _____ substance is made.
5. Damp air makes _____ form on iron.

B. Put ✓ for true and ✗ for false

1. ☐ Our body is also matter.
2. ☐ A gas has a fixed shape.
3. ☐ Paper burning is a chemical change.
4. ☐ Melting ice can never be reversed.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Stone	( ) Gas
2. Milk	( ) Chemical change
3. Air	( ) Solid
4. Egg cooking	( ) Liquid

D. Answer in short

1. Write the three states of matter with one example of each.

2. Write one difference between a physical and a chemical change.

3. How do we know that air takes up space?

E. Draw and label

Draw three boxes: solid, liquid and gas. In each box draw one thing and write its name.

FOR TEACHERS

Answer key

Chapter 5: Matter and Its Changes

Practice questions

1. (a) Anything that takes up space and has mass
2. (a) Air takes up space
3. (a) Solid
4. (b) Liquid
5. (c) Oxygen
6. (c) Paper burning
7. (c) Water freezing into ice
8. (a) A new substance is made
9. (a) Particles
10. False
11. True
12. False
13. True
14. False
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. matter · 2. solid · 3. container · 4. new · 5. rust

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Gas · (4) Chemical change · (1) Solid · (2) Liquid

D. Answer in short:

1. Solid (stone), liquid (water), gas (air).
2. No new substance is made in a physical change, but a new substance is made in a chemical change.
3. When we blow up a balloon, air fills it and it gets bigger.

Picture credits

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